

Employee Attrition and Annual Leave Predictor

Venura Gunaratne

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1 The logistic regression model highlights key drivers of employee attrition expressed as Odds Ratios (OR):

Term	Odds Ratio	2.5% CI	97.5% CI	pvalue
age	1.033	0.851	1.253	0.745
daily_rate	1.164	1.006	1.348	0.042
distance_from_home	0.621	0.534	0.721	0.000
hourly_rate	0.857	0.736	0.997	0.046
monthly_income	0.965	0.475	1.970	0.922
monthly_rate	0.878	0.758	1.016	0.082
num_companies_worked	0.605	0.511	0.712	0.000
percent_salary_hike	1.318	1.051	1.657	0.017
stock_option_level	1.488	1.216	1.825	0.000
total_working_years	2.959	2.078	4.262	0.000
years_at_company	0.252	0.176	0.358	0.000
years_in_current_role	1.967	1.500	2.592	0.000
years_since_last_promotion	0.644	0.523	0.790	0.000
years_with_curr_manager	2.006	1.552	2.607	0.000

Term	Odds Ratio	2.5% CI	97.5% CI	pvalue
environment_satisfaction_1	2.977	1.791	4.981	0.000
environment_satisfaction_2	1.568	1.000	2.467	0.051
environment_satisfaction_3	1.842	1.268	2.687	0.001
environment_satisfaction_4	NA	NA	NA	NA
job_satisfaction_1	3.018	1.794	5.102	0.000
job_satisfaction_2	1.844	1.152	2.958	0.011
job_satisfaction_3	1.762	1.207	2.579	0.003
job_satisfaction_4	NA	NA	NA	NA
marital_status_Single	0.673	0.557	0.811	0.000
marital_status_Divorced	1.264	1.060	1.511	0.010
over_time_Yes	0.314	0.270	0.365	0.000
relationship_satisfaction_1	1.064	0.632	1.788	0.816
relationship_satisfaction_2	0.584	0.360	0.945	0.029
relationship_satisfaction_3	1.072	0.728	1.579	0.723
relationship_satisfaction_4	NA	NA	NA	NA
work_life_balance_1	1.045	0.711	1.534	0.824
work_life_balance_2	0.822	0.608	1.110	0.201
work_life_balance_3	1.001	0.741	1.351	0.997
work_life_balance_4	NA	NA	NA	NA
business_travel_Travel_Frequently	0.657	0.574	0.750	0.000
business_travel_Non.Travel	1.591	1.315	1.941	0.000
department_Human.Resources	28.471	0.131	NA	0.963
department_Sales	1.537	0.641	3.922	0.353
education_1	0.765	0.309	1.914	0.564
education_2	0.968	0.348	2.717	0.951
education_3	0.942	0.470	1.899	0.868
education_4	0.928	0.645	1.336	0.689
education_5	NA	NA	NA	NA
education_field_Technical.Degree	1.267	0.847	1.912	0.253
education_field_Life.Sciences	2.796	1.433	5.579	0.003
education_field_Marketing	1.626	1.059	2.528	0.028
education_field_Medical	1.946	1.040	3.714	0.040

Term	Odds Ratio	2.5% CI	97.5% CI	pvalue
education_field_Other	1.715	1.207	2.459	0.003
gender_Female	1.441	1.242	1.676	0.000
job_involvement_1	2.090	1.336	3.302	0.001
job_involvement_2	1.210	0.856	1.722	0.284
job_involvement_3	1.422	1.046	1.940	0.026
job_involvement_4	NA	NA	NA	NA
job_level_1	0.001	0.000	0.006	0.000
job_level_2	0.000	0.000	0.001	0.000
job_level_3	0.001	0.000	0.010	0.000
job_level_4	0.038	0.014	0.092	0.000
job_level_5	NA	NA	NA	NA
job_role_Human.Resources	0.039	NA	4.972	0.961
job_role_Laboratory.Technician	0.460	0.326	0.643	0.000
job_role_Manager	1.057	0.726	1.556	0.775
job_role_Manufacturing.Director	0.683	0.529	0.874	0.003
job_role_Research.Director	1.638	1.118	2.487	0.015
job_role_Research.Scientist	0.770	0.536	1.097	0.151
job_role_Sales.Executive	0.275	0.111	0.641	0.004
job_role_Sales.Representative	0.485	0.289	0.792	0.005
performance_rating_X4	0.912	0.723	1.149	0.435

2 Predictors for annual leave frequency modeled via Poisson regression (Incident Rate Ratios):

Term	IRR	2.5% CI	97.5% CI	pvalue
age	1.000	0.993	1.006	0.894
daily_rate	1.000	1.000	1.000	0.872
distance_from_home	1.000	0.996	1.005	0.894
hourly_rate	1.000	0.998	1.002	0.907
monthly_income	1.000	1.000	1.000	0.010
monthly_rate	1.000	1.000	1.000	0.734
num_companies_worked	1.008	0.990	1.026	0.370
percent_salary_hike	1.003	0.986	1.020	0.745

Term	IRR	2.5% CI	97.5% CI	pvalue
stock_option_level	1.020	0.960	1.083	0.523
total_working_years	1.001	0.989	1.012	0.881
years_at_company	1.019	1.005	1.032	0.006
years_in_current_role	0.997	0.980	1.016	0.779
years_since_last_promotion	0.996	0.981	1.012	0.629
years_with_curr_manager	0.975	0.958	0.993	0.007
environment_satisfaction_1	1.018	0.683	1.519	0.929
environment_satisfaction_2	1.127	0.808	1.575	0.481
environment_satisfaction_3	1.021	0.813	1.281	0.856
environment_satisfaction_4	NA	NA	NA	NA
job_satisfaction_1	1.637	1.065	2.520	0.025
job_satisfaction_2	3.165	2.192	4.591	0.000
job_satisfaction_3	2.203	1.762	2.756	0.000
job_satisfaction_4	NA	NA	NA	NA
marital_status_Single	0.986	0.884	1.100	0.800
marital_status_Divorced	0.945	0.852	1.047	0.281
over_time_Yes	2.062	1.905	2.231	0.000
relationship_satisfaction_1	0.897	0.602	1.339	0.595
relationship_satisfaction_2	0.925	0.663	1.293	0.647
relationship_satisfaction_3	0.966	0.773	1.205	0.760
relationship_satisfaction_4	NA	NA	NA	NA
work_life_balance_1	1.242	0.769	1.998	0.372
work_life_balance_2	1.167	0.841	1.615	0.353
work_life_balance_3	1.096	0.878	1.366	0.418
work_life_balance_4	NA	NA	NA	NA
business_travel_Travel_Frequently	1.054	0.954	1.162	0.299
business_travel_Non.Travel	1.068	0.937	1.213	0.315
department_Human.Resources	1.030	0.618	1.652	0.906
department_Sales	1.031	0.725	1.456	0.862
education_1	1.027	0.366	2.795	0.959
education_2	1.097	0.456	2.596	0.834
education_3	1.040	0.615	1.748	0.883

Term	IRR	2.5% CI	97.5% CI	pvalue
education_4	0.999	0.780	1.280	0.994
education_5	NA	NA	NA	NA
education_field_Technical.Degree	1.469	1.008	2.166	0.048
education_field_Life.Sciences	1.598	1.119	2.314	0.011
education_field_Marketing	1.602	1.091	2.381	0.018
education_field_Medical	1.434	1.002	2.080	0.052
education_field_Other	1.601	1.094	2.368	0.017
gender_Female	0.981	0.906	1.062	0.633
job_involvement_1	0.826	0.502	1.350	0.449
job_involvement_2	0.992	0.705	1.391	0.965
job_involvement_3	0.982	0.793	1.217	0.869
job_involvement_4	NA	NA	NA	NA
job_level_1	2.304	0.622	8.617	0.213
job_level_2	1.321	0.410	4.332	0.644
job_level_3	1.178	0.527	2.665	0.691
job_level_4	1.064	0.744	1.523	0.735
job_level_5	NA	NA	NA	NA
job_role_Human.Resources	1.185	0.707	2.049	0.530
job_role_Laboratory.Technician	0.974	0.796	1.193	0.802
job_role_Manager	1.092	0.792	1.498	0.588
job_role_Manufacturing.Director	0.933	0.779	1.120	0.457
job_role_Research.Director	1.019	0.765	1.352	0.898
job_role_Research.Scientist	0.920	0.751	1.127	0.420
job_role_Sales.Executive	0.918	0.633	1.341	0.656
job_role_Sales.Representative	0.894	0.585	1.372	0.607
performance_rating_X4	1.042	0.881	1.233	0.632

3 RMSE and MAE

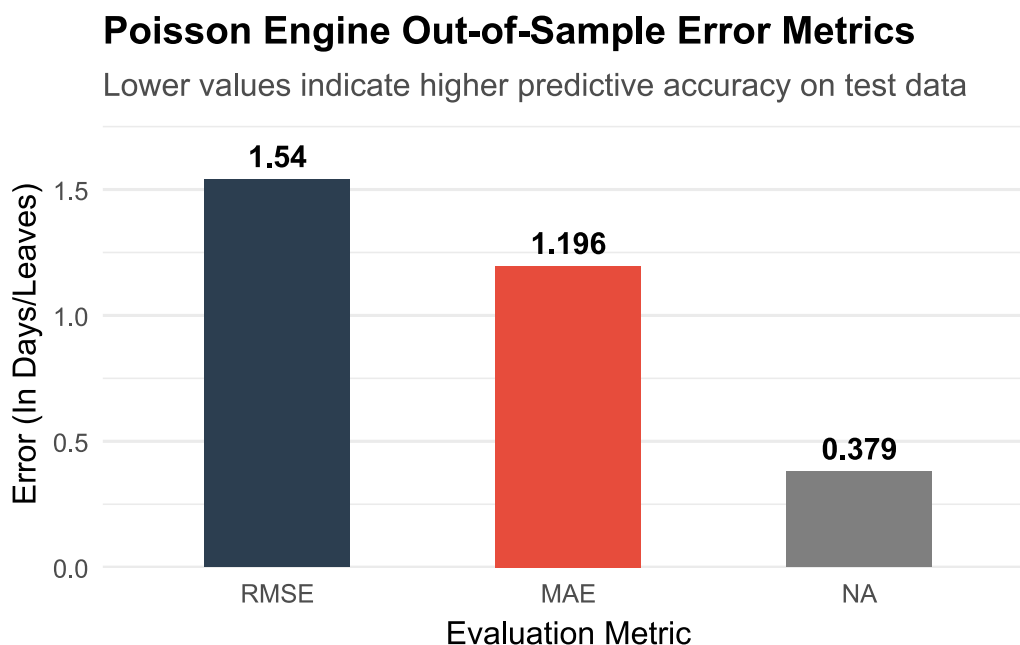


Figure 1: Out-of-Sample Performance Metrics for Poisson Leave Prediction Engine

4 Model Performance & Discrimination

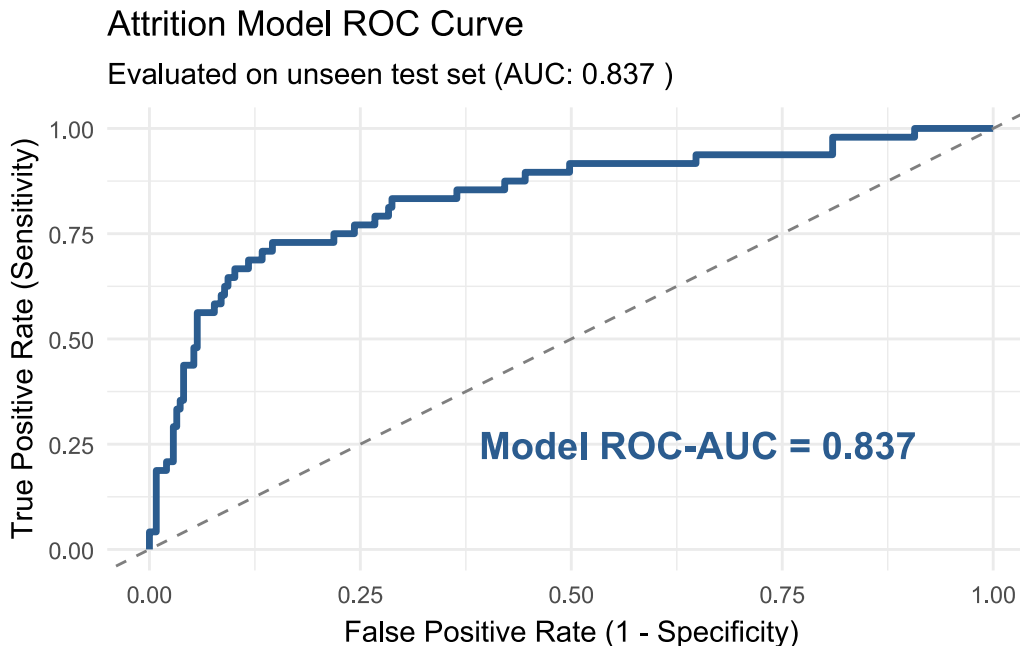


Figure 2: Model Performance and ROC-AUC

The model achieved an out-of-sample **ROC-AUC of 0.837**, demonstrating strong predictive capability in distinguishing high-risk attrition candidates from stable employees on unseen evaluation data.

5 Top 10 Predictors of Attrition

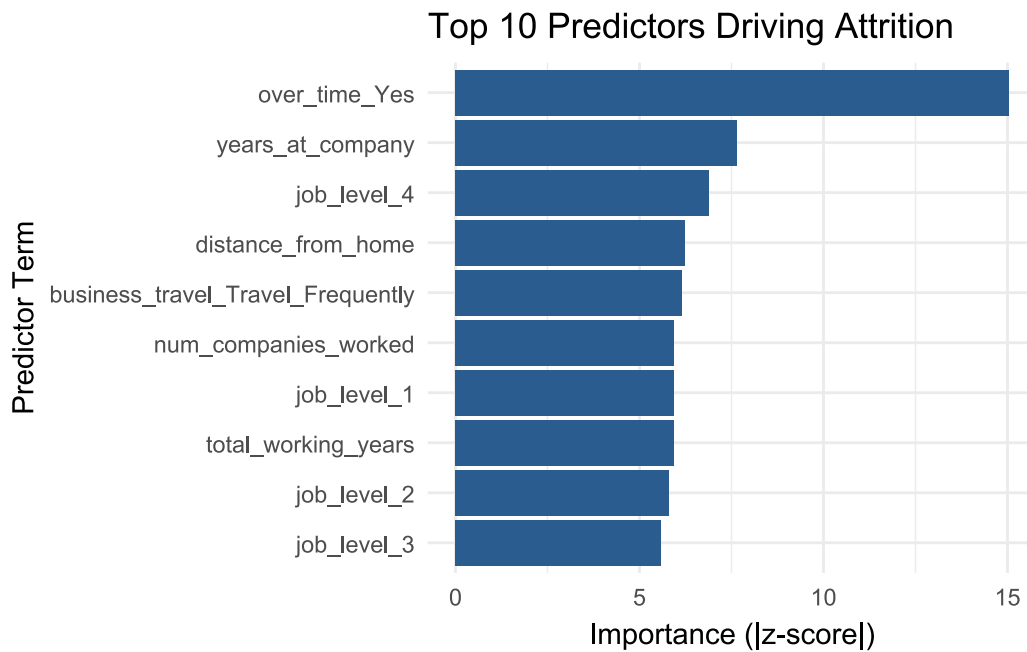


Figure 3: Top 10 Feature Importance Scores for Attrition Prediction

6 Comparison Between Log Regression and XGBoost Models

Model Architecture	Performance Metric	Test Score
Logistic Regression	ROC-AUC	0.837
XGBoost	ROC-AUC	0.801

Both baseline Logistic Regression and hyperparameter-tuned XGBoost were evaluated on an untouched 20% test data set (stratified by attrition status).

7 Comparative ROC Curve Between the Two Models

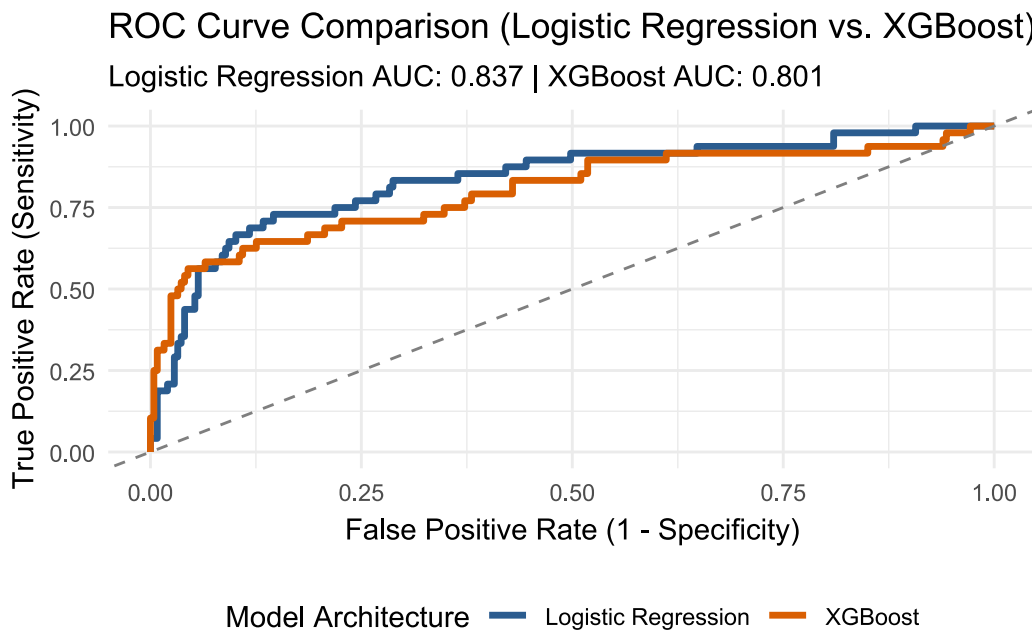


Figure 4: Combined ROC Curves for Attrition Prediction Models

8 Combined Multi-Model Comparison

Table 1

Model Engine	Target Outcome	Metric	Test Score
Logistic Regression	attrition	ROC-AUC	0.837
XGBoost	attrition	ROC-AUC	0.801
Poisson Regression	total_annual_leaves	RMSE	1.540
Poisson Regression	total_annual_leaves	RSQ	0.379
Poisson Regression	total_annual_leaves	MAE	1.196

Overall Predictive Engine Metric Comparison

ROC-AUC evaluated for Attrition models | RMSE & MAE evaluated

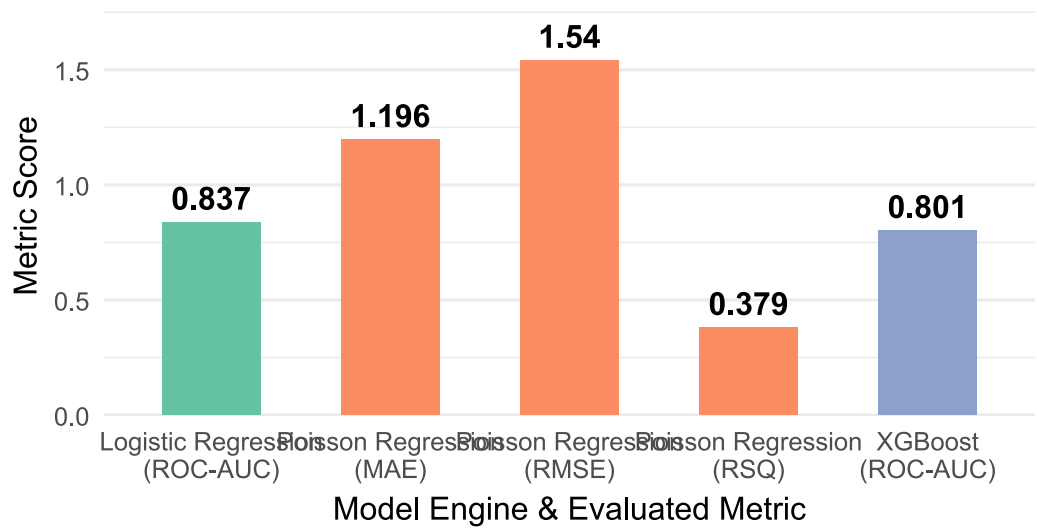


Figure 5: Comparative Overview of Model Metrics Across Prediction Engines

9 Monte Carlo Model Summary

Table 2

Simulated Outcome	Expected Mean	Lower 2.5% Bound	Upper 97.5% Bound
Total Resignations (Attrition)	472.7	447	498
Total Cumulative Leave Days	3339.6	3226	3452

10 Monte Carlo Distribution

Monte Carlo Attrition Risk Distribution

Dashed orange lines mark the 95% simulated interval bounds

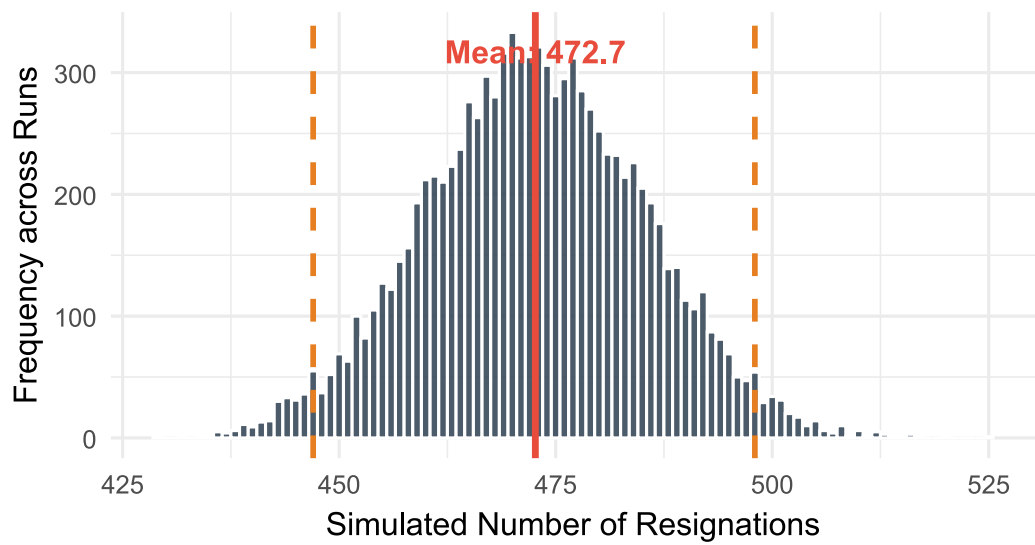


Figure 6: Simulated Organizational Attrition Distribution Across 10,000 Monte Carlo Futures